

Assignment 1 COMP 551

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Abstract. The assignment's objective consisted of implementing, training, and evaluating two distinct machine learning algorithms, K-Nearest Neighbours and Decision Trees. After loading the datasets, data entries were cleaned, encoded, scaled and standardized to ensure proper calculations for distances and avoid noisiness. Then, statistical analysis was performed on the features to assess their predictive power and interdependence, and to identify the most relevant variables. Following feature selection, each model was implemented through its own Python class containing training, testing, and predicting functions. Accuracy functions were also defined to evaluate each method's effectiveness with different hyperparameters. Once the setup was finalized, tests were run to determine which model was more suitable for each dataset. Ultimately, both models were equally accurate for the penguin dataset and KNN proved to be slightly more accurate (higher AUROC) than DT at predicting the presence of heart diseases.

1 Introduction

The K-Nearest Neighbors and Decision Trees models were trained on two datasets with the goal of predicting the presence of heart disease in patients and the type of penguin species, respectively. The heart disease data set accumulated health-related data from medical patients in the Cleveland area, such as blood sugar and maximum heart rate capacity. For the penguin dataset, it included body measurements from adult penguins, such as mass and flipper length.

Our experiments consisted of comparing the AUROC curves, inputting different K values and tree depth, trying alternative distance and cost functions, and ranking features according to predictive power. In particular, we used the Gower distance ¹ in the KNN to account for the mixed data types in the heart disease dataset. Gower distance is a method that involves sub-calculations depending on the nature of the feature. Let x_1 be a point in our data set, and let x_2 be the test point.

$$D_{\text{Gower}}(x_1, x_2) = \frac{1}{N} \sum_{i=1}^N s_i(x_1, x_2)$$

For dimension i of N :

Categorical features:

$$s_i(x_1, x_2) = \mathbb{I}(x_{1i} = x_{2i})$$

(since 1 if $x_{1i} \neq x_{2i}$, otherwise 0)

Numerical features:

$$s_i(x_1, x_2) = \frac{|x_{1i} - x_{2i}|}{\text{Range}_i}$$

Ordinal features:

$$s_i(x_1, x_2) = \frac{|x_{1i} - x_{2i}|}{\text{Max}_i}$$

¹Machine Learning Interview. (2023, February 15). Gower Distance for Mixed Data [Video]. YouTube. <https://www.youtube.com/watch?v=PHu8VoPv-o4>

In all cases, a value of 0 signifies a higher similarity. The Gower distance is the sum of all the i -th distances for $1 \leq i \leq N$.

2 Methods

Our study employs two machine learning techniques for predictive analysis:

Weighted K-Nearest Neighbor (KNN):

Weighted-KNN is a variation of the standard KNN algorithm that classifies data points based on their 'K' nearest neighbors in the training set. It operates under the premise that similar instances are likely to be in close proximity. The effectiveness of the algorithm depends on the choice of the parameter 'K' and the distance function used. In our case, we decided to use the Euclidean, Manhattan and Gower distance functions as our primary distance metrics to measure similarity between data points.

In a standard KNN algorithm, all 'K' neighbors contribute equally to the classification. However, in Weighted-KNN, the influence of each neighbor is adjusted according to its proximity to the query point. The weight function is typically defined as:

$$\text{Weight of a neighbor} = \frac{1}{\text{distance}}$$

By assigning different levels of influence to neighboring points based on their distances, Weighted-KNN enhances the prediction accuracy. Closer neighbors have a stronger influence on the classification outcome, reducing the impact of distant neighbors that may be less relevant.

Decision Trees (DT):

Decision Trees model decisions in a tree structure, where data is recursively split into subsets based on the most significant attribute at each node. Nodes represent decision points, and leaves represent the outcomes or class labels.

A key feature of Decision Trees is their ability to have varying depths, meaning the tree can be shallow or deep depending on how many splits occur before reaching the leaves. Shallow trees may underfit the data, failing to capture complex patterns, while deeper trees can capture more intricate relationships but may overfit by including noise or irrelevant details.

Another crucial aspect of Decision Trees is the use of cost functions to determine the optimal features and thresholds at each node. The goal is to maximize information gain or minimize the loss in the resulting subsets. Common cost functions used in our experiment are Gini-Index, Misclassification and Entropy.

To optimize the depth of the tree, a validation set can be used to test different depths and select the one that yields the best accuracy. This helps avoid both underfitting and overfitting.

3 Datasets

After uploading both datasets, rows with missing or incoherent values were removed and the sets were reindexed. In both datasets, numerical features were scaled using Scikit-Learn StandardScaler. Categorical variables with two levels were converted into binary values (0 and 1).

After dropping the island feature, the penguin dataset counted 4 numerical features, 1 binary feature, and the target. The squared differences in mean highlighted the higher predictive power of culmen depth, body mass, and flipper length. However, accuracy decreased when the length of the culmen was removed and increased when the sex was dropped, so the four numerical features were included in the model.

The heart disease set initially had 6 numerical features, 4 categorical features, 3 binary features, and the target (i.e.: mixed data types). The target column included integers from 0 to 4. All non-zero values were converted into 1, which signified the presence of a heart

disease. Categorical variables with more than two levels were encoded using Scikit-Learn OneHotEncoder, increasing the total number of features to 21. Computing the squared difference of mean and correlation between each feature and the response column (with flexible thresholds of 0.1 and 0.3, respectively) discarded many irrelevant features. Finally, computing the Pearson correlation between each feature allowed us to limit multicollinearity by avoiding the presence of related features in the dataset (when two relevant features were related, the one with the lowest correlation with the target was dropped). Performing filter-based feature elimination on the heart disease dataset helped increase the accuracy of the KNN model to 0.88 at $K = 7$.

4 Results

To investigate how the choice of K impacts KNN’s performance, we tested multiple values of K (1–10) and recorded accuracy on both the validation and test datasets. For Dataset 1, the validation accuracy remained stable and nearly perfect across all K values, as shown in Figure 1. However, the test accuracy exhibited a slight downward trend, though the values remained relatively consistent and close to perfect. The optimal K was found at $K = 3$, where the training accuracy reached 97%, and the validation accuracy achieved a perfect 100%. ”For Dataset 2, the validation accuracy increased with the number of K , peaking at $K = 7$, where the corresponding test accuracy was 84.5%. The test accuracy fluctuated slightly up or down but remained relatively stable overall.

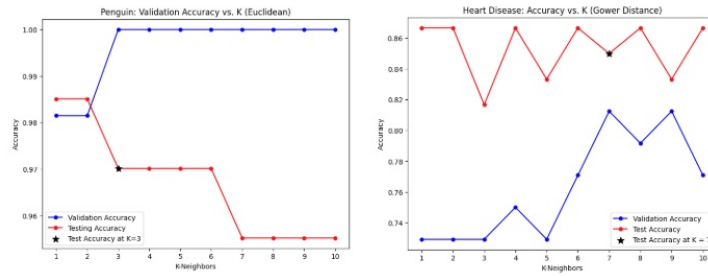


Figure 1: Penguin (left) & Heart Disease (right): Test accuracy vs. K

The maximum depth of a Decision Tree (DT) is an essential hyperparameter that significantly influences the model’s complexity and its generalization capabilities. To understand the effects of this parameter, we analyzed the DT’s performance on the Penguin and Heart Disease datasets by varying the maximum depth and for different cost functions. As shown in Figure 2, for Dataset 1, the accuracy initially increases and then stabilizes as the maximum tree depth increases. The best depth was determined based on the highest validation accuracy, which peaked at a depth of 2 using the misclassification cost, resulting in a test accuracy of 95%. For Dataset 2, the accuracies fluctuated but remained relatively stable. The highest validation accuracy occurred at depth = 5 when using the entropy cost, resulting in a test accuracy of 80% however, this did not correspond to the highest test accuracy.

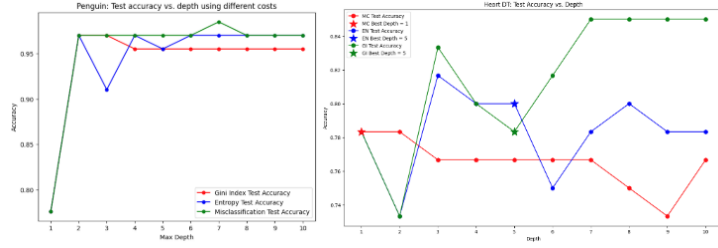


Figure 2: Penguin (left) & Heart Disease (right): Test accuracy vs. Depth

Weighted-KNN achieves high performance on the Penguin dataset, with 98.5% test accuracy and an AUROC of 0.98, regardless of whether the Euclidean or Manhattan distance is used as the cost function. This suggests that KNN is a highly effective model for the Penguin dataset, with minimal impact from the choice of distance metric at this particular $K = 3$.

The Decision Tree model performs well on the Penguin dataset, with accuracy ranging from 94.0% to 95.2%, depending on the cost function used. The Misclassification cost function results in the highest test accuracy (95.2%) and a high AUROC (0.96), which indicates that it slightly outperforms both the Gini Index and Entropy cost functions for this task.

Weighted-KNN on the Heart Disease dataset has a test accuracy of 85.0% and an AUROC of 0.91, which is fairly good but lower compared to its performance on the Penguin dataset. The choice of the Gower cost function is used for the Heart Disease dataset, possibly due to the nature of the dataset's features.

The Decision Tree shows a noticeable drop in performance on the Heart Disease dataset when compared to the Penguin dataset. The best accuracy (80.0%) occurs with the Entropy cost function at Depth = 5, but the AUROC remains relatively low (around 0.77–0.84). The Misclassification cost function at Depth = 1 results in a similar test accuracy (78.3%) with an AUROC of 0.84, indicating that deeper trees might not always lead to better results on this dataset.

4.1 Tables and Figures

Dataset	Model	Cost Function	Hyperparam.	Test Acc.	Test AUROC
Penguin	KNN	Euclidean	$K = 3$	98.5%	0.98
Penguin	KNN	Manhattan	$K = 3$	98.5%	0.98
Penguin	Decision Tree	Gini Index	Depth = 2	94.0%	0.95
Penguin	Decision Tree	Entropy	Depth = 2	94.3%	0.96
Penguin	Decision Tree	Misclassification	Depth = 2	95.2%	0.96
Heart Disease	KNN	Gower	$K = 7$	85.0%	0.91
Heart Disease	Decision Tree	Gini Index	Depth = 5	78.3%	0.78
Heart Disease	Decision Tree	Entropy	Depth = 5	80.0%	0.77
Heart Disease	Decision Tree	Misclassification	Depth = 1	78.3%	0.84

Table 1: Comparison of Models and Hyperparameters Across Datasets.

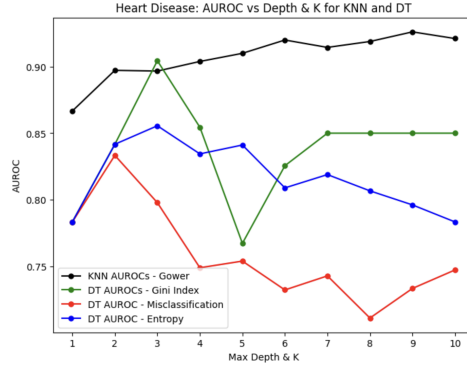


Figure 3: Heart Disease: AUROC vs Depth & K

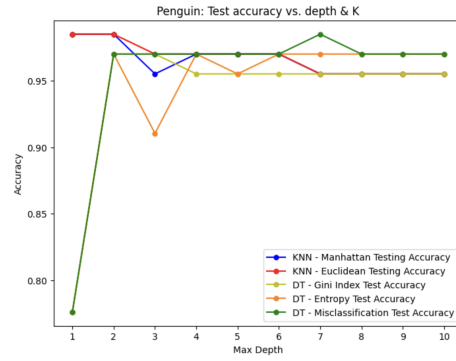


Figure 4: Penguin: Test Accuracy vs. Depth & K

Figure 5: Comparison of AUROC and Test Accuracy for KNN and Decision Trees.

We compared the feature importance scores from the Decision Tree method with the squared mean differences we calculated earlier, and the rankings showed some clear differences. In the Penguins dataset, both methods identified *culmen_length_mm*, *culmen_depth_mm*, and *flipper_length_mm* as important, but their rankings varied. The Decision Tree prioritized *culmen_length_mm*, while the squared mean difference ranked *culmen_depth_mm* highest. *Body_mass_g* was important in the squared mean difference method but wasn't used for splits in the Decision Tree.

For the Heart Disease dataset, *oldpeak* was important in both methods, but *age* was highly ranked by the Decision Tree and not in the squared mean difference approach. Meanwhile, *thal* and *ca*, which had the highest squared mean differences, weren't considered important by the tree. These differences occur because Decision Trees choose features that create the best splits, while the squared mean difference method measures class separation without considering interactions. Correlated features can also cause variations, as trees might rely on one feature while ignoring another. Overall, this shows that different methods can lead to different rankings depending on how they define importance.

5 Conclusion

In summary, the performance of our Weighted-KNN and Decision Tree (DT) models was more pronounced on Dataset 1 (penguins) compared to Dataset 2 (heart disease). Both models exhibited comparable accuracy on Dataset 1; however, on Dataset 2, the Weighted-KNN model demonstrated superior accuracy relative to the DT model. For further investigation, identifying and selecting the most relevant features for Dataset 2 could help improve the results.

6 Statement of contribution

Leen Assaf: Dataset Preprocessing, Running Experiments and Write-up
Vincent Barsalou: Dataset Preprocessing, Running Experiments and Write-up
Karl Wehbe: Model Implementation, Running Experiments, and Write-up